Represent Rational Numbers on the Coordinate Plane

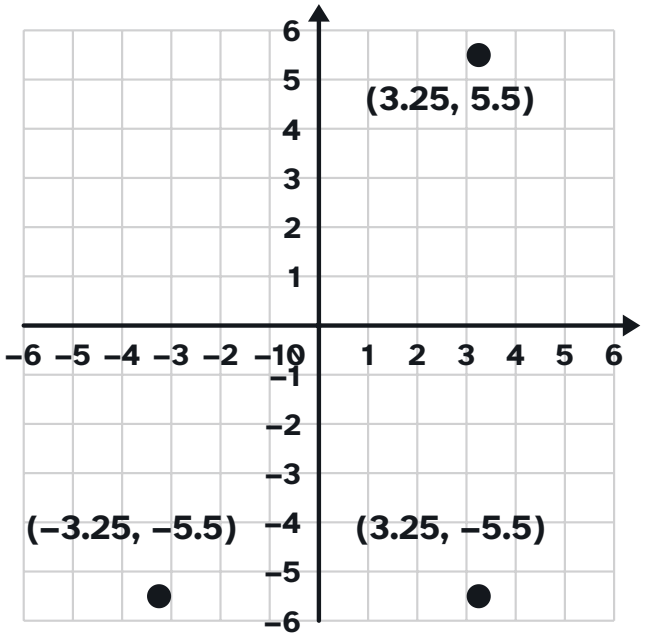
Flip a point across an axis and name where it lands.

Name _____ **Date** _____

Tonight's spark

A class mural plan puts a star at R, (3.25, –5.5), on a grid. The painter flips a copy of the star across the x-axis, like a mirror. Where does the copy land?

Guess first: does the copy land above or below the star?



Our class model: the mirror line

Try it

Work it out on paper. Write one answer on each line.

1 **Which coordinate changes sign when you reflect across the x-axis?**

A. The x-coordinate **B.** The y-coordinate **C.** Both coordinates

My answer _____

2 **Reflect (4, 3) across the x-axis.**

A. (–4, 3) **B.** (4, –3) **C.** (–4, –3) **D.** (4, 3)

My answer _____

3 **Reflect (6, 1) across the y-axis.**

A. (6, –1) **B.** (–6, –1) **C.** (1, 6) **D.** (–6, 1)

My answer _____

4 **Reflect (–5, 2) across the x-axis.**

A. (–5, –2) **B.** (5, –2) **C.** (5, 2) **D.** (–5, 2)

My answer _____

5 **Reflect (1.5, –4) across the y-axis.**

A. (1.5, 4) **B.** (–1.5, 4) **C.** (–1.5, –4) **D.** (1.5, –4)

My answer _____

Talk about it

Which coordinate changes sign when you reflect across the x-axis? Which stays the same?

Home mission

Lay a ruler on the table as the x-axis. Put a coin below it for the mural star. Ask: where does its mirror copy go? Put a second coin there. Then compare how far each coin is from the ruler. Which number changed, the left-right one or the up-down one?

If we get stuck, ask

- Which axis are we reflecting across?
- Does that axis run left-right or up-down? Which value has to change?
- Is the image the same distance from the mirror line?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/7-5-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it